Kilian Pouderoux

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Final-year Microengineering Master's student at EPFL & CentraleSupélec, seeking a full-time position starting in June 2027. Expertise in microsystems, optics, and nanotechnology applied to high-tech, fast-paced industrial or research projects.

Professional experience

Coherent – **Master's thesis** Zürich, CH / 2026 (6 months)

Master's thesis on the experimental investigation of grating properties in high-power DFB lasers, combining optical setup development, device characterization, and simulation to establish robust laser design rules

- Design and build an automated free-space and fiber-based optical setup for the electro-optical characterization of laser devices
- Develop data analysis pipelines to process, visualize, and validate key laser performance metrics
- Simulate DFB laser behavior and compare it with measurements to assess grating and epitaxial design variations

IMEC - **Microfluidics & photonics intern** Leuven, BE / 2026 (6 months)

Interdisciplinary research internship combining microfluidics, photonics, and systems engineering to develop an integrated on-chip cell sorting platform for biomedical diagnostics applications

- Design and assemble an integrated optical and microfluidic setup, combining fluidic, electronic, and photonic components
- Develop data acquisition and processing pipelines to analyze and visualize cell sorting performance metrics
- Prepare biological samples and conduct wet lab experiments to characterize the chip's sorting performance

Education

École Polytechnique Fédérale de Lausanne, EPFL Lausanne, CH / 2024 –

Master's degree in Microengineering, minor in Computational Science and Engineering

Courses : optics, photonics, MEMS, AI, nanotechnology, microfabrication, HPC, quantum computing and simulations, advanced numerical analysis, stochastic simulation, structural biology.

CentraleSupélec – Université Paris-Saclay Paris, FR / 2022 - 2024

Engineering curriculum

Courses : mathematics, engineering, energy, finance, cloud computing, quantum physics, nuclear engineering, control systems, optimization, numerical modeling, signal processing.

École Polytechnique Fédérale de Lausanne, EPFL Lausanne, CH / 2019 - 2022

Bachelor's degree in Mechanical Engineering

Academic projects

- Laboratory of Applied Photonic Devices (LAPD), EPFL : development of a compact multi plane spatial light modulation setup for optical information processing, using Pytorch and high-performance computing.
- Laboratory of Hybrid Photonics (HYLAB), EPFL : Electrode optimization for high-Q resonators using field-based surface analysis and CST simulations.
- Development of an autonomous mobile robot : implementation of ML algorithms on embedded systems with computer vision for human identification and target following.

Associative experience

EPFL Spacecraft Team* – *Team leader ground segment Lausanne, CH / 2024 - 2026

Lead a team of 8 to develop and operate ground stations for satellite communications. Developed a full-stack software for distributed control of antennas and mission control.

Forum CentraleSupélec* – *Event and quality manager Paris, FR / 2022 - 2024

Organized a major business forum with 200+ companies and a €1.3 million turnover. Ensured compliance with ISO standards and managing relations and logistics with contractors. Developed automation algorithms to improve team efficiency.

Skills and interests

Languages	French : native English : C1 Italian : A2
Programming	Python – C/C++ – Git – Bash – CUDA – MPI
Software	Matlab – LabVIEW – Catia – Arduino/Raspberry Pi – Docker
Sports	Basketball Tennis Ski Kitesurf Foil
Interests	Astronomy Science Mythology Travel

References

Christophe Moser : Professor, Laboratory of Applied Photonics Devices at EPFL : christophe.moser@epfl.ch

Christophe Salzmann : Professor, Automatic Control Laboratory at EPFL : christophe.salzmann@epfl.ch